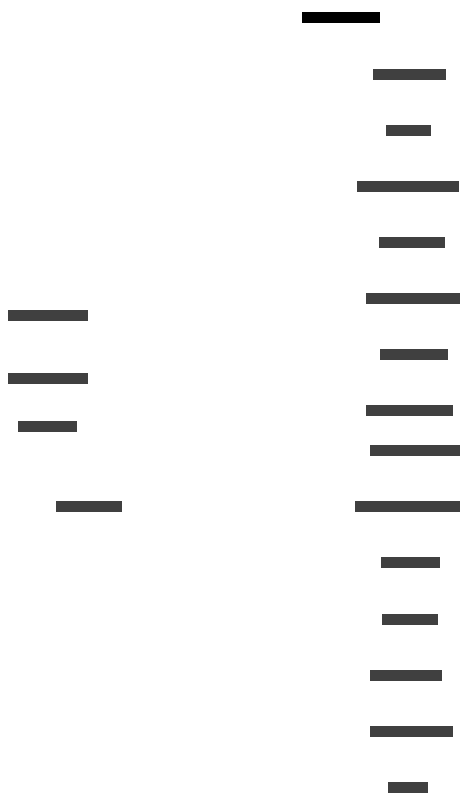


CRAFT — Pitch Package

1. What CRAFT Is

An audited, retrieval-augmented AI assistant for CSA mission engineering and finance. This codebase MVPs the seams. Nothing here ships yet — the pattern shipping is the point.

2. Architecture



L1 Context

CRAFT is a Flask + Dash app. Every request enters through a route-gated RBAC layer (see L1, L2). A LangGraph primary agent runs a bounded ReAct loop, binds tools via a `Tool` protocol, and emits `AGENT_ACTION` events after every step. Storage sits behind a `WorkspaceStore` protocol; external corpora sit behind a `DataSource` protocol; connectors line up along the perimeter. The route-gated + audit spine gives the system its safety guarantees — every request checked for auth + role before reaching business logic, every agent step logged with trace IDs for end-to-end reconstruction.

L2 Container

Component views: - **L3 Trio** — agent internals (LangGraph ReAct graph, tool wrappers, audit envelope) · tools & connectors (Tool + DataSource protocols, perimeter connectors) · route + auth (RBAC gates, `_ROUTE_RULES`, test anchors). - **UC Trio** — Risk id + HITL (UC-E2) · Parametric cost + HITL (UC-F1) · Vendor cost + SAP connector (UC-F4).

3. Requirements → Design

Capability	Req IDs	Chosen dep / pattern	Blocker	Ready when...	UC
Primary agent	UMR-001/011/013/016/018/019/094/095, AAR-001/003/006, ASG-005	LangGraph ReAct graph · ChatProvider protocol · <code>recursion_limit=25</code>	none	ChatProvider seam; sub-agent = new graph node	E2, F1, F4
Retrieval + RAG	UMR-002/003/004/005/006/009/010/060/067, ARR-002-007	Azure AI Search (hybrid vector+BM25) · DataSource + Tool protocols	IT: Azure AI Search deploy; data: source register	new HybridRetrievalTool on Tool protocol	E2, F1
Audit + traceability	UMR-015/027/045/061/062/093, AAR-005, ASG-003	AgentAction envelope · AgentActionLog protocol · JSON lines → Azure Log Analytics	IT: Log Analytics workspace + immutability policy	AgentActionLog swappable sink	E2, F1, F4
HITL approvals	UMR-021-026/035/039/044/049, HITL-001-007, ASG-002	LangGraph interrupt primitive · Postgres approval queue · Dash reviewer UI	data: CSA risk-tier taxonomy	ChatProvider + interrupt seam; queue = new table	E2, F1
RBAC + data scoping	UMR-012/017/058/065/066, AAR-002/007	Route-Gated RBAC (<code>_ROUTE_RULES</code> , <code>_ROLE_LEVEL</code>) + SqliteDataStore scoping + <code>QueryTool.set_user</code>	none for perimeter; per-tool RBAC pending	<code>_ROUTE_RULES</code> list appends; tool-permission column additive	E2, F1, F4
Session + workspace (V3)	UMR-014, UMR-092, UMR-095	Split WorkspaceStore into persistent SqliteStore + new SessionScratchStore (ephemeral)	design: session-scoped agent lifecycle	WorkspaceStore protocol; scratch class = drop-in	dream
Guardrails + containment	UMR-020/029/030, ASG-001/004/011	Azure AI Content Safety · APIM rate limit · Azure Container Apps sandbox (UMR-093)	procurement: Content Safety + Container Apps	tool wrapper adds moderation call; sandbox = new Tool class	dream
Connectors + external systems	UMR-051-055	DataSource protocol + MSAL (Entra ID / SharePoint) · Flask REST · SAP / STK / MATLAB planned	procurement: SAP integration path; licenses: STK/ MATLAB	each connector = new DataSource class + <code>services.py</code> line	F4
Domain capabilities	UMR-007/008/031-034/036-038/040-043/046-048/050/075, ARR-001		data: CSA risk	uniform pattern — see § 4	E2, F1

Capability	Req IDs	Chosen dep / pattern	Blocker	Ready when...	UC
		Each domain = new Tool impl over existing seams (retrieval, HITL, audit, RBAC)	taxonomy, historical mission DB; procurement: Translator for reports		
Deployment, SLA, config	UMR-028/056/057/059/064/069/070/074/085, ASG-006	Azure App Service (PBMM Canadian region) · APIM + App Gateway (WAF) · SSC LaunchPad HA target · git-versioned config	IT: PBMM landing zone; SSC LaunchPad readiness	env-var config; migrations = numbered SQL	dream

4. Use Cases

UC-E2 · Risk identification + HITL (Engineering). A mission engineer uploads a CADRe Part A document. The primary agent retrieves relevant risk sections, runs a keyword + LLM classifier, flags risks — then hits a HITL gate. A reviewer approves or rejects; approved risks write to the risk register (planned). [View state model →](#)

UC-F1 · Parametric cost estimation + HITL (Finance). A program admin requests a cost estimate for a new mission profile. The agent retrieves 5 most-similar historical missions, computes a weighted Euclidean distance, presents the estimate — then hits a HITL gate before commitment. [View state model →](#)

UC-F4 · Vendor cost roll-up + SAP connector. A finance analyst queries vendor costs across past programs. The agent routes through a planned SAP connector implementing DataSource, extracts vendor line items, aggregates, and audits. Blocked on SAP integration — the DataSource seam is proven with LocalFileSource. [View state model →](#)

Other flows use the same spine. Parametric budget scenarios, vendor-cost aggregation across missions, anomaly triage, bilingual report drafting, cross-mission telemetry compare, cost-per-requirement decomposition — each lands as a new Tool implementation plus (where needed) a new DataSource. Retrieval, RBAC scoping, HITL gates, AGENT_ACTION audit envelope: unchanged. No new architecture. Timeline gated by the blockers in § 5.

5. What's Blocked, What We're Ready For

- **Azure AI Search Serverless deploy** (IT + procurement) — unlocks hybrid retrieval, citations, faithfulness scoring. Seam: DataSource + new HybridRetrievalTool.
- **Azure Log Analytics workspace + immutability policy** (IT) — unlocks tamper-proof audit at UMR-027 grade. Seam: AgentActionLog swappable sink.
- **Azure AI Content Safety + Container Apps sandbox** (procurement) — unlocks UMR-020 guardrails + UMR-093 sandboxed code exec. Seam: tool wrapper moderation, new CodeExecTool.
- **Entra ID app registration + SharePoint / SAP connectors** (IT + finance team) — unlocks CSA-native corpus + cost data. Seam: each = new DataSource class.
- **CSA risk taxonomy + risk register schema** (CSA domain) — unlocks UMR-036–040 and the HITL flows around risk writes.
- **Historical mission DB + parametric feature schema** (CSA finance / mission engineering) — unlocks UMR-031–035 cost estimation.
- **SSC LaunchPad availability** (SSC) — unlocks HA topology for UMR-057.

6. System-Health Tooling

Tool	Purpose	Runs where
ruff	Lint (replaces flake8, isort, pyupgrade)	pre-commit + CI
pyright	Strict type checking, no # type: ignore	pre-commit + CI
pytest + pytest-cov	156 tests, ~91% coverage, ≥80% threshold enforced	pre-commit + CI
bandit	Static security scan	pre-commit
pip-audit	CVE audit of installed deps	pre-commit
detect-secrets	Prevent committing credentials	pre-commit

`app/tests/test_architecture.py` auto-discovers feature folders and enforces: no cross-feature internal imports (protocol boundaries), required files per feature, root file whitelist. `app/tests/test_rules.py` adds: logging coverage audit, no hardcoded secrets, function length caps, import completeness. Same suite runs `.pre-commit-config.yaml` and `.github/workflows/ci.yaml`. New invariant = new test. Local warning + CI gate, no extra infra.